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SOURCE: MSC.194(80).pdf
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED-17 -

Regulation 7
Attained subdivision index A

1 The attained subdivision index A is obtained by the summation of the partial indices As, Ap and Al, (weighted as shown) calculated for the draughts ds, dp and dl defined in regulation 2 in accordance with the following formula:

A = 0.4 As + 0.4Ap + 0.2Al

Each partial index is a summation of contri butions from all damage cases taken in consideration, using the following formula:

A = pi si

where:

i represents each compartment or group of compartments under consideration,

pi accounts for the probability that only the compartment or group of compartments under consideration may be flooded, disregarding any horizontal subdivision, as defined in regulation 7-1,

si accounts for the probability of survival after flooding the compartment or group of compartments under consideration, and includes the effect of any horizontal subdivision, as defined in regulation 7-2.

2 In the calculation of A, the level trim shall be used for the deepest subdivision

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group of compartments under consideration, and includes the effect of any horizontal subdivision, as defined in regulation 7-2.

2 In the calculation of A, the level trim shall be used for the deepest subdivision draught and the partial subdivisi on draught. The actual service trim shall be used for the light service draught. If in any service conditio n, the trim variation in comparison with the calculated trim is greater than 0.5% of Ls, one or more additional calculations of A are to be submitted for the same draughts but different trims so that, for all service conditions, the difference in trim in compar ison with the reference trim used for one calculation will be less than 0.5% of Ls.

3 When determining the positive righting lever (GZ) of the residual stability curve, the displacement used should be that of th e intact condition. Th at is, the constant displacement method of calculation should be used.

4 The summation indicated by the above formula shall be taken over the ship's subdivision length (Ls) for all cases of flooding in which a single compartment or two or

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displacement method of calculation should be used.

4 The summation indicated by the above formula shall be taken over the ship's subdivision length (Ls) for all cases of flooding in which a single compartment or two or more adjacent compartments are involved. In the case of unsymmetrical arrangements, the calculated A value should be the mean value obt ained from calculations involving both sides. Alternatively, it should be take n as that corresponding to the side which evidently gives the least favourable result.

5 Wherever wing compartments are fitted, contribution to the summation indicated by the formula shall be taken for all cases of flooding in which wing compartments are involved. Additionally, cases of simultaneou s flooding of a wing compartment or group of compartments and the adjacent inboard compartment or group of compartments, but excluding damage of transverse extent greater than one half of the ship breadth B, may be added. For the purpose of this regulation, transverse extent is measured inboard from ship's side, at right angle to the centreline at the level of the deepest subdivision draught.

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added. For the purpose of this regulation, transverse extent is measured inboard from ship's side, at right angle to the centreline at the level of the deepest subdivision draught.

RESOLUTION MSC.194(80)
(adopted on 20 May 2005)
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6 In the flooding calculati ons carried out according to the regulations, only one breach of the hull and only one free surface ne ed to be assumed. The assumed vertical extent of damage is to extend from the baseline upwards to any watertight horizontal subdivision above the waterline or higher. However, if a lesser ex tent of damage will give a more severe result, such extent is to be assumed.

7 If pipes, ducts or tunnels are situated within the assumed extent of damage, arrangements are to be made to ensure that progressive flooding cannot thereby extend to compartments other than those assumed flooded. However, the Administration may permit minor progressive flooding if it is dem onstrated that its effects can be easily controlled and the safety of the ship is not impaired.

Regulation 7-1
Calculation of the factor pi

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permit minor progressive flooding if it is dem onstrated that its effects can be easily controlled and the safety of the ship is not impaired.

Regulation 7-1
Calculation of the factor pi

1 The factor pi for a compartment or group of comp artments shall be calculated in accordance with paragraphs 1.1 and 1.2 using the following notations:

j = the aftmost damage zone number involved in the damage starting with No.1 at the stern;

n = the number of adjacent damage zones involved in the damage;

k = is the number of a partic ular longitudinal bulkhead as barrier for transverse penetration in a damage zone counted from shell towards the centre line. The shell has k = 0;

x1 = the distance from the aft terminal of Ls to the aft end of the zone in question;

x2 = the distance from the aft terminal of Ls to the forward end of the zone in question;

b = the mean transverse distance in metr es measured at right angles to the centreline at the deepest subdivision loadline between the shell and an assumed vertical plane extended betw een the longitudinal limits used in calculating the factor pi and which is a tangent t o, or common with, all or

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centreline at the deepest subdivision loadline between the shell and an assumed vertical plane extended betw een the longitudinal limits used in calculating the factor pi and which is a tangent t o, or common with, all or part of the outermost portion of the longitudinal bulkhead under consideration. This verti cal plane shall be so orientated that the mean transverse distance to the shell is a maximum, but not more than twice the least distance between the plane and th e shell. If the upper part of a longitudinal bulkhead is below the deepest subdivision loadline the vertical plane used for determination of b is assumed to extend upwards to the deepest subdivision waterline. In any case, b is not to be taken greater than B/2.

If the damage involves a single zone only:

pi = p(x1j,x2j) [r(x1j,x2j,bk) - r(x1j,x2j,bk-1)]

If the damage involves two adjacent zones:

pi = p(x1j,x2j+1) [r(x1j,x2j+1,bk) - r(x1j,x2j+1,bk-1)]
- p(x1j,x2j) [r(x1j,x2j,bk) - r(x1j,x2j,bk-1)]
- p(x1j+1,x2j+1) [r(x1j+1,x2j+1,bk) - r(x1j+1,x2j+1,bk-1)]
RESOLUTION MSC.194(80)

(adopted on 20 May 2005)

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SOURCE: MSC.194(80).pdf
pi = p(x1j,x2j+1) [r(x1j,x2j+1,bk) - r(x1j,x2j+1,bk-1)]
- p(x1j,x2j) [r(x1j,x2j,bk) - r(x1j,x2j,bk-1)]
- p(x1j+1,x2j+1) [r(x1j+1,x2j+1,bk) - r(x1j+1,x2j+1,bk-1)]
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(adopted on 20 May 2005)
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If the damage involves three or more adjacent zones:

pi = p(x1j,x2j+n-1) [r(x1j,x2j+n-1,bk) - r(x1j,x2j+n-1,bk-1)]
- p(x1j,x2j+n-2) [r(x1j,x2j+n-2,bk) - r(x1j,x2j+n-2,bk-1)]
- p(x1j+1,x2j+n-1) [r(x1j+1,x2j+n-1,bk) - r(x1j+1,x2j+n-1,bk-1)]
+ p(x1j+1,x2j+n-2) [r(x1j+1,x2j+n-2,bk) - r(x1j+1,x2j+n-2,bk-1)]

and where r(x1, x2, b0) = 0

1.1 The factor p(x1, x2) is to be calculated according to the following formulae:

Overall normalized max damage length: Jmax = 10/33
Knuckle point in the distribution: Jkn = 5/33